SHORTS

THE WEB WE LOST

The author takes a nostalgic look at some pre-web 2.0, pre-social media heavy days of the internet and enlists some features that the current generation of web users will never know of. It is a good walk down memory lane.
http://dgit.in/T5WMIL

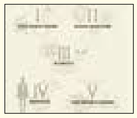
VIRAL PHOTO PROJECTS OF 2012

Wired has come up with this wonderful compilation of the most engaging viral photo projects which include the Haunted House series, Sea Creature series and so on. A visual treat!
http://dgit.in/SMGCpC

LIFE OF PI EDITOR INTERVIEWED

Here Tim Squyres, Life of Pi's film editor talks about the challenges posed while editing 3D films, making the CG tiger look realistic, if digital actors can replace human actors. If you haven't seen the film, please do!
http://dgit.in/ZghXN8

THE END?



This series from Verge takes a look at various conspiracy theories and has interviews with renowned theorists who touch upon topics such as the new world order, solar cataclysm, etc., which they sat will lead to doomsday.
http://dgit.in/V6MtF4

Probing The Matrix: Is our universe simulated, and if so… by who?

- By John Hewitt

PCMAG DIGITAL NETWORK

Interpreting the universe as a computer simulation is perhaps the inevitable byproduct of living in the computer age. The question today is not whether we live inside a simulation, but rather — what does it want, and what compromises with regard to the welfare its inhabitants might have been made to get it? The first rudimentary experiments to poke it in the belly and see how it jiggles are just getting underway.

If it's possible to hack the universe, then particle physicists must be its early phone phreakers. Fears, which turned out to exaggerated, were raised few years ago that a black hole might unwittingly be created inside the Large Hadron Collider. Arguably, it is more accurate to say that with the theories that prescribe the existence of subatomic particles that exist nowhere else in the known universe, these folks actually invent them, rather than just discover them. The fact that they then go out and literally create subatomic particles inside of these huge, specially built accelerators, raises a very real question — what is a subatomic particle? Is it the blip on the oscilloscope, or the whole accelerator itself?

THE MATRIX

High-energy astrophysicists study cosmic rays — the most energetic particles naturally produced by the universe, as opposed to high-energy particles in accelerators. They are believed to come from distant events, like supernova explosions, pulsars, and quasars, and are detected in the vicinity of Earth using satellite- or balloon-borne instruments. Scientists have accumulated a wealth of data about cosmic rays. It is therefore not surprising that a recent proposal entitled Constraints on the Universe as a Numerical Simulation [PDF][http://goo.gl/OWQAN] seeks to reveal the structure of the grid underlying the simulated universe, using measurements of the energy distribution of these particles.

After considerable theorizing, the authors determine that the distribution and direction of incoming cosmic rays at the highest energies can be used to derive a limit on the spacing of the grid. That is actually quite a bold statement. Fundamental limits to how small space can be divided have arisen in physics discussions before, particularly in reference to the Planck length, which is quantum mechanical in nature. Here the authors only consider a simulation of a very small region of space, on the order of just a few femtometers, and assume ordinary classical computation comprises the simulation — which is to say there is no quantum computing necessary.

Before any definite conclusions can be drawn from these observations, more complete measurements of cosmic rays need to be made for the simple reason that astrophysicists have yet to observe a firm upper bound to the cosmic ray energy spectrum — its "tail" dips above a certain energy, but it is still long and of unknown length. In other words, the more we continue to look for cosmic rays, the more we continue to find particles at ever higher energies, making plausible explanations for their existence all the more difficult.

If the universe is a simulation, why is it being simulated in the first place?

One thing to bear in mind is that what constitutes a "proof" under one set of assumptions may not be accepted as such under another. Mathematicians can convincingly show for example, that the square root of two is an irrational number by proving that the opposite idea is false. As Kurt Godel showed in his incompleteness theorem, undecidable statements can and will arise within any finite set of axioms. Depending on our particular axioms, the question of whether we live in a simulation could be provable, disprovable, or neither.

Assuming that an upper limit to cosmic ray energy is found or other methods of verification are uncovered, and the simulation is to some extent quantified, is it possible to ascertain the purpose of the simulation? If it might be said that we are carrying out some kind of computation — what computation? It is no great leap to suppose that any